RW–11121

Sub. Code
731201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Education

TEACHER BEHAVIOUR

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) "Nature of human behaviour determines the teacher and Learner behaviour" – Substantiate.

Or

(b) Write an essay about the acceptable behaviour of teachers required for present day education systems.

2. (a) List down various teacher strategies in handling deviant classroom situations.

Or

(b) Explain how for classroom interaction facilitate learning. Illustrate your answer.

3. (a) What is the significance of human relations? Outline the factors influencing human relations.

Or

(b) What are the roles of teachers in organisation and Communication process of an educational system?

4. (a) Outline the multi-dimensional roles of school-community relationship for the welfare of society.

Or

(b) What you consider as professional ethics and code of conduct for teachers as per New education policy? Explain.

5. (a) List down the competencies of master trainers. Explain how will you develop them among teachers?

Or

(b) Write a brief account about various researches in teacher behaviour in India and abroad.

RW–11068

Sub. Code
761201

M.Phil. DEGREE EXAMINATION, MAY 2016.

English

INDIAN ENGLISH LITERATURE

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) How does Ezekiel illustrate his view by comparing a poet to a lover and birdwatcher in "Poet, Lover, Birdwatcher"?

Or

(b) How far is Yeats statement true that "to read one line of Tagore is to forget all the troubles of the world"?

2. (a) Discuss Mahatma Gandhi's message of love and spirituality.

Or

(b) Examine the renaissance that Sri Aurobindo highlights in *The Renaissance in India.*

3. (a) How does Dattani deal with partition in *Final Solutions?*

Or

(b) Analyze the trial scenes in Tendulkar's *Silence*! *The Court is in Session.*

4. (a) Comment on the use of Time and Space in Ghosh's *The Shadow Lines.*

Or

(b) Show how satire is effectively used in *The God of Small Things.*

5. (a) Discuss the gradual stages of Saleem's character in *Midnight's Children.*

Or

(b) How does Bharati Mukherjee fictionalize the problems of immigration, expatriation, and cross-cultural assimilation in *Wife*?

RW–11069

Sub. Code
581201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Physics

MATERIAL SCIENCE OF THIN FILMS

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Explain the film thickness uniformity.

Or

(b) Illustrate the mechanism of electron beam evaporation method. Give its advantages, disadvantages and applications.

2. (a) Discuss in detail about the thermodynamics of chemical vapour deposition.

Or

(b) Describe the theory of plasma enhanced chemical vapour deposition process.

3. (a) Narrate the structure zone models for evaporated and sputtered coatings.

Or

(b) Briefly explain the Monte Carlo simulations.

4. (a) Explain the optical method for measuring film thickness.

Or

(b) Discuss in detail about the transmission electron microscopy.

5. (a) Describe the mechanism of auger electron spectroscopy with suitable diagram.

Or

(b) With neat sketches explain X-ray photoelectron spectroscopy.

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RW–11071

Sub. Code
581203

M.Phil. DEGREE EXAMINATION, APRIL 2016

Physics

CRYSTAL GROWTH AND CHARACTERISATION

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

($5 \times 15 = 75$)

1. (a) Define Nucleation? Explain the critical size of homogenous and heterogeneous nuclei and the rate of homogenous nucleation.

Or

(b) Discuss in detail about the BCF theory. Explain the growth rate on diffuse interface.

2. (a) Write a short notes on

(i) Distillation

(ii) Sublimation

(iii) Precipitation.

Or

(b) With neat sketch explain gas chromatography. Give its application.

3. (a) Briefly explain Bridgman — Stockbarger technique

Or

(b) Write in detail about the condensation, sputtering and growth by irreversible reactions.

4. (a) How a crystal can be grown in aqueous solution method. Explain its various techniques.

Or

(b) Narrate the gel growth method. Mention its advantages, disadvantages and its application.

5. (a) Describe the observation of defects in crystals using optical microscopy.

Or

(b) How do you analyze the thermal properties of a crystal using TG/DTA?

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RW–11072

Sub. Code
571101

M.Phil. DEGREE EXAMINATION, APRIL 2016

Mathematics

RESEARCH METHODOLOGY FOR ALGEBRA

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. $(5 \times 15 = 75)$

All questions carry equal marks.

1. (a) (i) Define tower and refinement. Let G be a finite group. Prove that an abelian tower of G admits a cyclic refinement.

(ii) Let H be a normal subgroup of a group G. Prove that G is solvable if and only if H and $\frac{G}{H}$ are solvable.

Or

(b) (i) If $n \geq 5$, prove that the alternating group A_n is simple.

(ii) Prove that a group of order 35 is cyclic.

2. (a) (i) Construct the Grothendieck group $K(M)$ of a commutative monoid. If the cancellation law holds in M, prove that the canonical map $\gamma : \mathrm{M} \to \mathrm{K(M)}$ is injective.

(ii) If $A = B \times C$ is a finite abelian group, prove that $A^{\wedge} \cong B^{\wedge} \times C^{\wedge}$. Deduce that a finite abelian group is isomorphic to its own dual.

Or

(b) (i) Show that the completion and the inverse limit $\underline{\lim} \frac{G}{H}$ are isomorphic under natural mappings.

(ii) Let A and B be two groups whose set-theonetic intersection is {1}. Prove that there exists a group. $A \circ B$ containing A, B as subgroups such that $A \cap B = \{1\}$ along with the property : Every element $\neq$ of $A \circ B$ has a unique expression as a product $a_1 \ldots a_n (n \geq 1, a_i \neq 1 \forall_i)$ with $a_i \in A$ or $a_i \in B$ and such that if $a_i \in A$, then $a_{i+1} \in B$ and if $a_i \in B$, then $a_{i+1} \in A$.

3. (a) (i) Prove that every ideal is contained in some maximal ideal.

(ii) State and prove Chinese Remainder theorem.

Or

(b) (i) Prove that every principal entire ring is factorial.

(ii) Describe the construction of $S^{-1}A$.

4. (a) (i) Prove that a sequence $0 \to y' \to y \to y''$ is exact if and only if

$0 \to Hom_A(x, y') \to Hom_A(x, y) \to Hom_A(x, y'')$ is exact for all x.

(ii) Prove that $Hom_A(M \oplus M', N) \cong Hom_A(M, N) \times Hom_A(M', N)$ as abelian groups.

Or

(b) (i) Define abelian category. Give an example.

(ii) Let V be a vector space over K and let W be a subspace. Prove that $\dim_K V = \dim_K W + \dim_K \frac{V}{W}$.

5. (a) (i) Let F be a free module, and M a submodule. Prove that M is free and its dimension is less than or equal to the dimension of F.

(ii) State and prove Snake lemma.

Or

(b) (i) Prove that the direct limits exist in the category of modules over a ring.

(ii) Assume that (A_n) satisfies ML. Given an exact sequence $O \to (A_n) \to (B_n) \xrightarrow{g} (C_n) \to O$ of inverse systems, prove that

$O \to \underline{\lim} A_n \to \underline{\lim} B_n \to \underline{\lim} C_n \to O$ is exact.

RW–11073

Sub. Code
571102

M.Phil. DEGREE EXAMINATION, APRIL 2016

Mathematics

FUNCTIONAL ANALYSIS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** the questions.

All questions carry equal marks.

$(5 \times 15 = 75)$

1. (a) (i) Prove that every topological vector space is a Hausdorff space.

(ii) Prove that every topological vector space has a balanced convex local base.

Or

(b) (i) Prove that L^p is a locally bounded F-space.

(ii) Prove that a topological vector space X is normable if and only if its origin has a convex bounded neighbourhood.

2. (a) State and prove Banach-Steinhaus theorem.

Or

(b) State and prove closed graph theorem.

3. (a) (i) If X is a locally convex space then prove that X^* separates points on X.

(ii) If f is a continuous linear functional on a subspace M of a locally convex space X, then there exists $\wedge \in X^*$ such that $\wedge = f$ on M.

Or

(b) State and prove Krein-Milman theorem.

4. (a) (i) Let M be a closed subspace of a Banach space X. The Hahn-Banach theorems extends each $m^* \in M^*$ to a functional $x^* \in X^*$. Define $\sigma m^* = x^* + M^{\perp}$. Then prove that σ is an isomorphism of M^* onto $X^* / M^{\perp}$.

(ii) Prove that every finite dimensional normal space is reflexive.

Or

(b) If X and Y are Banach spaces and $T \in \mathfrak{B}(X, Y)$. Then prove that T is compact if and only if T^* is compact.

5. (a) (i) Prove that every differentiable operator D^{α} is a continuous mapping of $D(\Omega)$ into $D(\Omega)$.

(ii) If $\wedge_i \to \wedge$ in $D'(\Omega)$ and $g_i \to g$ in $C^{\infty}(\Omega)$ then prove that $g_i \wedge_i \to g \wedge$ in $D'(\Omega)$.

Or

(b) Suppose $\{h_j\}$ is an approximate identity on R^n, $\varphi \in D$ and $u \in D'$. Then prove that :

(i) $\lim_{j \to \infty} \varphi * h_j = \varphi \in D$

(ii) $\lim_{j \to \infty} u * h_j = u \in D'$.

RW–11074

Sub. Code
571203

M.Phil. DEGREE EXAMINATION, APRIL 2016

Mathematics

Elective — MEASURE THEORY

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** the questions.

$(5 \times 15 = 75)$

1. (a) Suppose M is σ algebra in X, and Y is a topological space. Let f map X into Y. The prove the following

(i) If Ω is the collection of all sets $E \subset Y$ such that $f^{-1}(E) \in M$, then Ω is a σ algebra in Y.

(ii) If f is measurable and E is a Borel set in Y, then $f^{-1}(E) \in M$.

(iii) If $y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \in M$ for every real α, then f is measurable.

(iv) If f is measurable, if Z is a topological space, if $g : Y \to Z$ is a Borel mapping, and if $h = g \circ f$, then $h = X \to Z$ is measurable.

Or

(b) (i) Let u and v be real measurable functions on a measurable space X, let ϕ be a continuous mapping of the plane into a topological space Y, and define $h(x) = \phi(u(x), v(x))$ for $x \in X$. Prove that $h : X \to Z$ is measurable.

(ii) Suppose $f : X \to [0, \infty]$ is measurable, and $\phi(E) = \int_E f \, d\mu$, then prove that ϕ is a measure on M, and $\int_X g \, d\phi = \int_X gf \, d\mu$ for every measurable on X with range in $[0, \infty]$.

2. (a) (i) State and prove Lusin's theorem.

(ii) If $\{K_\alpha\}$ is a collection of compact subsets of a Hausdorff space and if $\bigcap_\alpha K_\alpha = \phi$, show that some finite sub collection of $\{K_\alpha\}$ also has empty intersection.

Or

(b) (i) Let X be a locally compact Hausdorff space in which every open set is σ compact. Let λ be any positive Borel measure on X such that $\lambda(K) < \infty$ for every compact set K. Then prove that λ is regular.

(ii) State and prove Vitali – Carathedory theorem.

3. (a) (i) Show that $L^p(\mu)$ is a complete metric space, for $1 \le p \le \infty$ and for every positive measure μ.

(ii) State and prove Holders inequality.

Or

(b) (i) Let p and q be conjugate exponents, $1 < p < \infty$. Let X be a measure space, with measure μ. Let f and g be measurable functions on X, with range in $[0, \infty]$. Prove that $\int_X fg\, d\mu \le \left\{\int_X f^p d\mu\right\}^{\frac{1}{p}} \left\{\int_X g^q d\mu\right\}^{\frac{1}{q}}$ and $\left\{\int_X (f+g)^p d\mu\right\}^{\frac{1}{q}} \le \left\{\int_X f^p d\mu\right\}^{\frac{1}{p}} + \left\{\int_X g^p d\mu\right\}^{\frac{1}{q}}$.

(ii) State and prove Minkowski's inequality.

4. (a) State and prove Lebesgue – Radon – Nikodym theorem.

Or

(b) (i) If X is a locally compact Hausdorff space, prove that every bounded linear functional ϕ on $C_0(X)$ is represented by a unique regular complex Borel measure μ, in the sense that $\phi f = \int_X f\, d\mu$.

(ii) State and prove Hahn decomposition theorem.

5. (a) If $f : [a,b] \to R^1$ is differentiable at every point of $[a,b]$ and $f^1 \in L^1$ on $[a, b]$, prove that $f(x) - f(a) = \int_a^x f'(t)\, dt\, (a \le x \le b)$.

Or

(b) (i) If $f \in L^1(R^k)$, show that almost every $x \in R^k$ is a Lebesgue point of f.

(ii) State and prove the fundamental theorem of Calculus.

RW–11075

Sub. Code
571102

M.Phil. DEGREE EXAMINATION, APRIL 2016

Mathematics

FUNCTIONAL ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) (i) Prove that every topological vector space is a Hamsdorft space.

(ii) Justify that infinite-dimensional topological vector space is locally compact.

Or

(b) (i) Define: Seminorm on a vector space X. Show that in every locally convex space exists a separating family of continuous seminorms.

(ii) Show that $C^{\infty}(\Omega)$ is a Fve'chet space with the Heine-Borel property.

2. (a) (i) State and prove category theorem.

(ii) State and prove the uniform boundedness principle.

Or

(b) (i) State and prove the closed graph theorem.

(ii) Prove that a bilinear mapping is continuous if it is continuous at the origin.

3. (a) (i) State and prove the separation theorem.

(ii) Define: Weak topology of a topological vector space. Also prove that the weak closive E_w^- of a convex subset E of a locally convex space is equal to its orginal closure $\bar{E}$

Or

(b) (i) State and prove Banach-Aloglu theorem.

(ii) Show that the convex hull of a compact set in IR^n is compact.

4. (a) (i) state and prove any one of the closed range theorem.

(ii) Define: Annillators $M^{\perp}$ and ${}^{\perp}N$, where M is a subspace of a Banach space X and N is a subspace of the normed dual of X. Also describe the duality between these two types of annihilators.

Or

(b) (i) Define: Compact operator. Illustrate.

(ii) State and prove any two properties of the spectrum of compact $T \in B(x)$.

5. (a) (i) State Pick-Nevanlinna interpolation problem. Show that if $\{z_0, z_{1`}\} = \left\{0_1, \frac{1}{2}\right\}$ and $\{w_0, w_1\} = \left\{0, \frac{2}{3}\right\}$ this problem admits no solution.

(ii) Show that if Pick-Nevanilnna problem has solutions then it contains a finite Blaschke product.

Or

(b) (i) Define: Hear measure on compact Groups.

(ii) State and prove Kakutani's fixed point theorem.

RW–11076

Sub. Code
561551

M.Phil. DEGREE EXAMINATION, APRIL 2016

Computer Science

INFORMATION AND NETWORK SECURITY

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

Each question carries 15 marks.

1. (a) Explain the various classical encryption methods in detail.

Or

(b) Explain the

(i) Auto-key cipher (ii) Cryptographic codes.

2. (a) Explain the various types of attacks with example.

Or

(b) Describe the Elliptic Curve Cryptography algorithm with example.

3. (a) Explain the Real-time stegnography with an example.

Or

(b) Explain the standard Hashing algorithms.

4. (a) Describe Entity Authentication techniques.

Or

(b) Explain the technological issues in (i) Face Recognition (ii) Fingerprint Recognition.

5. (a) Explain (i) SSL operations (ii) Virus and Worms.

Or

(b) Explain the various types of Firewalls in detail.

RW–11077

Sub. Code
519201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Zoology

ANIMAL BIOTECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Discuss in detail about the scope of animal biotechnology.

Or

(b) Discuss about the advanced molecular techniques in disease diagnosis.

2. (a) Describe the characteristic features of cloning vectors in prokaryotic and eukaryotic systems.

Or

(b) Explain the principle and procedure of nuclear transplantation.

3. (a) Describe the pros and cons of recombinant DNA technology.

Or

(b) Explain about the various methods of DNA sequencing.

4. (a) What are the various gene transfer methods available for eukaryotes?

Or

(b) Discuss in detail about animal genomics and proteomics.

5. (a) Discuss in detail about the pros and cons of stem cell technology.

Or

(b) Explain the applications of stem cell technology.

RW–11078

Sub. Code
921201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Library and Information Science

AREA OF SPECIALIZATION – USER STUDIES

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

Each question carries 15 marks.

(5 × 15 = 75)

1. (a) Define information seeking behaviour. Explain the need for and types of information seeking pattern.

Or

(b) Write in detail some of the major user studies conducted in India.

2. (a) Discuss the information needs of different users of Higher education information system.

Or

(b) Critically evaluate the factors affecting the information needs of special library users.

3. (a) Explain the role of international organisations in conducting user surveys.

Or

(b) How do you evolve a theory of user behaviour? Mention the limitations in behaviour research.

4. (a) Trace the origin and development of user education.

Or

(b) Write short notes on :

(i) Failure Analysis

(ii) Orientation to freshers.

5. (a) Explain the parameters used to evaluate user studies.

Or

(b) Discuss the impact of internet and electronic resources on information seeking behaviour.

RW–11079

Sub. Code
507201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Nanoscience and Technology

NANOMATERIALS FOR POWER SOURCES, MAGNETIC DEVICES AND THEIR TOXICITY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Explain in detail about :
   (i) III-V
   (ii) II-VI solar cells.

   Or

   (b) Discuss in detail about :
   (i) Nanosolar cells
   (ii) Photocatalytic and photobiological generation of hydrogen fuel.

2. (a) Discuss in detail about Lithium, Ni-Cd and Metal hydride battery.

   Or

   (b) Explain in detail about fabrication and the advantage of various types of supercapacitors.

3. (a) Explain in detail how Methanol and Microbial fuel cell works and its advantages.

   Or

   (b) Describe in detail about the principle and application of PEM and Solid oxide fuel cells.

4. (a) Explain in detail about principles and advantages of nanomagnetic behaviour and magnetic reversal.

Or

(b) Explain in detail about :

(i) Magnetic recording

(ii) Nanodisk and nanoring memories and

(iii) Magnetic switches.

5. (a) Write an essay about nanohazard and nanotoxicity with precautionary assessment.

Or

(b) Write short notes :

(i) Cytotoxicity

(ii) Environmental effects and

(iii) Evaluation of consumer exposure to nanomaterials.

RW–11080

Sub. Code
507204

M.Phil. DEGREE EXAMINATION, APRIL 2016

Nanoscience and Technology

SYNTHETIC ROUTES OF NOVEL NANO-BIOMATERIALS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Answers **all** questions.

(5 × 15 = 75)

1. (a) Explain in detail about :

   (i) Fluoroimmunoassays

   (ii) Quantum dot-Encoded beads.

Or

   (b) Discuss in detail about :

   (i) Quantum dot labeling cell surface

   (ii) Quantum dot imaging the angiotensin.

2. (a) Discuss in detail about gold nanoparticle conjugates and their use in intracellular imaging and gold nanoshell conjugate antibodies.

Or

   (b) Explain in detail how surface functionalized nanoparticles are used for controlled drug delivery and nanocluster synthesis.

3. (a) Explain in detail about nanostructured DNA templates and its applications.

Or

(b) Describe in detail about the synthetic nanoscale elements for delivery of materials and real time cell dynamics.

4. (a) Explain in detail about Nature's nanotechnology with suitable case studies.

Or

(b) Explain in detail about :

(i) science at the biology,

(ii) scales at the bio-nano interface.

5. (a) Write an essay about Bionanocomposite synthesis, applications and milestones.

Or

(b) Write short notes :

(i) Metal oxide

(ii) Proteins

(iii) DNA and RNA polymer based and

(iv) Core-shell structured nanocomposites.

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RW–11115

Sub. Code
661102/ 642102/ 681102

M.Phil. DEGREE EXAMINATION, APRIL 2016

TECHNIQUES OF BUSINESS RESEARCH/STATISTICAL TECHNIQUES OF RESEARCH/TECHNIQUES OF RESEARCH

(Common for Commerce/Management/Bank Management)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(7 \times 3 = 21)$

Answer **all** questions.

All questions carry equal marks.

1. What are the properties of a good measure of central tendency?

2. Distinguish between correlation and auto-correlation.

3. What purpose does testing for difference between means serve?

4. What is the significance of Scheffe's test?

5. What is fit index?

6. What is canonical correlation?

7. What is a hypothesis?

Section B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

8. Differentiate between qualitative and quantitative research methods with suitable examples.

9. Find out the co-efficient of correlation from the following data :

X	65	66	67	67	68	69	71	73
Y	67	68	64	68	72	70	69	70

10. State the assumptions made in analysis of variance.

11. The details regarding the monthly sales performance two salesmen A and B given below. Which salesman is more consistent in sales performance and why? Sales performance for 10 months.

Average in Lakh Rs.	A	B
S.D	12.2	16.4
	1.7	3.8

12. Write down the applications of cluster analysis.

13. Explain the use of Auto-regressive Conditional heteroscedasticity (ARCH).

Section C (2 × 10 = 20)

Answer **all** questions.

All questions carry equal marks.

14. (a) What are the characteristics of mean, mode and median? When and why each of these averages can be used?

Or

(b) One thousand girls in a college were graded according to their I.Q. and the economic conditions of their homes. Use chi-square test to find out whether there is any association between economic condition at home and I.Q. of girls (Given $V = 1; \chi^2_{0.05} = 3.84$).

	I.Q.		
Economic conditions	High	Low	Total
Rich	100	300	400
Poor	350	250	600
Total	450	550	1000

15. (a) Explain clearly the concept of a factor analysis suggesting a suitable example. What are its uses?

Or

(b) Explain conjoint analysis with a suitable example. State its uses.

Section D (1 × 10 = 10)

(Compulsory)

16. The following are the weight gains (in pounds) of two random samples of young Turkeys fed two different diets but otherwise kept under identical conditions.

Diet 1:	16.3	10.1	10.7	13.5	14.9	11.8	14.3	10.2
Diet 2:	21.3	23.8	15.4	19.6	12.0	13.9	18.8	19.2
Diet 1:	12.0	14.7	23.6	15.1	14.5	18.4	13.2	14.0
Diet 2:	15.3	20.1	14.8	18.9	20.7	21.1	15.8	16.2

Use the Mann-Whitney U test at the 0.01 level of significance to test the null hypothesis that the two populations sampled have identical distributions against the alternative hypothesis that on the average the second diet produces a greater gain in weight.

RW–11116

Sub. Code
661201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Commerce

RESEARCH FOR BUSINESS DECISIONS

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (7 × 3 = 21)

Answer **all** questions.

All questions carry equal marks.

1. What is policy research?

2. Write a note on new product development research.

3. List down the factors to be considered for the conduct of research on leadership.

4. What do you mean by strategic alliance?

5. Mention the reasons for the conduct of research on global financial institutions.

6. State the determinants of capital structure.

7. Outline the importance of social responsibility research.

Section B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

8. Bring out in detail the various dimensions of business research.

9. Discuss the factors to be considered for conducting new product development research.

10. Explain the significance of research in personnel decision making.

11. Describe the different issues to be addressed while conducting research for mergers and acquisitions.

12. How will you analyse the reactions of investors during financial recession?

13. The analysis of working capital helps in the effectiveness of business management – Explain.

Section C (2 × 10 = 20)

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

14. (a) Examine the various types of research and their relative merits and demerits.

Or

(b) Identify the major problems in conducting Marketing Research in India. Suggest possible solutions to overcome these problems.

15. (a) Discuss in detail the applications of business strategy research in the context of mergers and acquisitions.

Or

(b) Describe the utility of research on credit management for banks and financial institutions.

Section D (1 × 10 = 10)

Compulsory question

16. A detergent manufacturer has approached you for preparing a printed advertisement copy for marketing his product. How will you support your work with marketing research?

RW–11119

Sub. Code
642203

M.Phil. (Management) DEGREE EXAMINATION, APRIL 2016

ADVANCED MARKETING MANAGEMENT

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section I (7 × 3 = 21)

Answer **all** questions.

1. What is meant by 'targeting and positioning'?

2. What is 'portfolio matrices'?

3. What is the relationship between pricing and break-even analysis?

4. What do you understand by the term 'disintermediation'?

5. What is meant by 'advertisement copy'?

6. What is 'mega marketing'?

7. What do you mean by 'Launch mistakes'?

Section II (4 × 6 = 24)

Answer any **four** questions.

8. Specify the market defender strategies available to service firms.

9. Bring out the reasons for the failure of a new product, with an example.

10. Distinguish between Leader pricing and Predatory price cutting.

11. Explain the factors influencing the modem retailing.

12. Explain the components of sales promotion strategies used by retailers?

13. Explain Database marketing with an example.

Section III (2 × 10 = 20)

Answer **all** questions.

14. (a) Apply PEST analysis as a tool and conduct pest analysis of any ONE of the following:

    (i) Internet based Business,

    (ii) computer lap-top marketing.

    Or

    (b) There is no fundamental differences between marketing of industrial goods and consumer goods-comment.

15. (a) Explore the relationship between marketing and supply chain management with a suitable example.

    Or

    (b) What are the requisites for the success of direct marketing?

Section IV (1 × 10 = 10)

(Compulsory)

16. Read the following case and answer the questions.

    Soori Company is manufacturing and marketing toys and furniture for Play School children in North India. At present, they want to enter South Indian Market. The product is to be branded and positioned in the market. If you are the Market Researcher and this assignment is given to you for a market study,

    (a) How will you plan the Market Research?

    (b) How will you segment and target the product?

RW–11120

Sub. Code
642204

M.Phil. DEGREE EXAMINATION, APRIL 2016

Management

STRATEGIC HUMAN RESOURCE MANAGEMENT

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (7 × 3 = 21)

Answer **all** questions.

All questions carry equal marks.

1. What do you mean by benchmarking in SHRM?
2. What is emotional intelligence?
3. How would you relate job evaluation to KPA?
4. Distinguish between counseling and mentoring.
5. What are the limitations of HR audit?
6. What do you mean by 'breaking the glass ceiling'?
7. What is career plateau?

Part B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

8. Differentiate between the strategies that are practiced in traditional HRM and SHRM.
9. What factors determine success in HR outsourcing?

10. Critically evaluate the benefits that are provided by any one of the leading information technology companies to combat job hopping.

11. Describe the role of training methods which contributes to organizational learning.

12. Brief the methods to improve quality of work life.

13. What are the factors that influence organization commitment?

Part C $(2 \times 10 = 20)$

Answer **all** questions.

All questions carry equal marks.

14. (a) Explain the process of succession planning.

Or

(b) Discuss different approaches to knowledge management.

15. (a) Imagine you are the Manager-Business process re-engineering in a leading organization. What are your roles and responsibilities?

Or

(b) Discuss the issues related to OD in virtual learning organizations.

Part D $(1 \times 10 = 10)$

(Compulsory)

16. Read the following case and answer the questions :

OTIS Elevator Company had concerns that its old paper-based performance appraisal system was too slow and cumbersome. There were also concerns whether the raters could be assured of the confidentiality to their

rating. Because of these problems, the company wanted a better system for appraising and development the performance of its engineering managers. Specially, the company was interested in enhancing these managers' project management and project team leadership skills, The engineering managers needed substantial improvement in their skills, and the company wanted a performance appraisal system that would provide feedback from the managers' subordinates, peers and customers as well as their direct superiors.

Given these concern, it is not surprising that OTIS Elevator decided to develop a 360–degree feedback system. The innovative aspect of the company's approach to the 360-degree system is that the company decided to base the system on the internet and its own intranet. An independent contractor, E-Group developed the system and handles the collation and analysis of the feedback information.

E-group chose a 75-item survey called LEAPS. which measures seven dimensions of leadership. for the 360-degree instrument. The instrument was loaded on a website so that all raters can pull up the information and complete the appraisal in approximately 20 minutes. After completing the appraisal, they simply submit the results via e-mail to E-group to process. Because the system is encrypted, the company is able to provide greater confidentiality and anonymity for the raters than with the previous paper-and-pencil system. In addition to the LEAPS item, the company included a fairly large set of other item to assess managers technical competency and their contributions to the business. E-group was able to provide appraisal profiles for the managers within three days after the last of the evaluators e-mailed their input for the manager. In addition, the profile of actual ratings for each manager from E-group also includes an ideal leadership profile developed by OTIS executives. By comparing his actual ratings with the ideal profile. managers can identify areas for future development.

OTIS Elevator chose to use the system only for developmental purposes, although recently it began to consider other purposes for the system.

Questions :

(a) How was the 360-degree appraisal better than the traditional appraisal system in OTIS Elevators?

(b) What problems do you think OTIS Elevator is likely to experience in this process?

RW–11067

Sub. Code
421201

M.Phil. DEGREE EXAMINATION, APRIL 2016

Tamil

தமிழ் உரை மரபுகள்

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

அனைத்து வினாக்களுக்கும் கட்டுரை வடிவில் விடை எழுதுக.

$(5 \times 15 = 75)$

1. (அ) நன்னூல் குறிப்பிடும் உரை உத்திகளை விளக்குக.

(அல்லது)

(ஆ) உரைகள் தமிழுக்குச் செய்த பங்களிப்புகளை மதிப்பீடு செய்க.

2. (அ) சேனாவரையரின் தொல்காப்பிய உரைத்திறனை மதிப்பீடு செய்க.

(அல்லது)

(ஆ) பேராசிரியர் தொல்காப்பிய அதிகார அமைப்புப் பெயரும் பற்றிக் கூறும் கருத்துக்களைத் தொகுத்துத் தருக.

3. (அ) பிற உரையாசிரியர்களின் கருத்துக்களை நச்சினார்க்கினியர் மறுக்கும் முறையை விளக்குக.

(அல்லது)

(ஆ) நச்சினார்க்கினியரின் பத்துப்பாட்டு உரைவழி அறியலாகும் சங்ககாலச் சமுதாயத்தின் பண்பாட்டை விளக்குக.

4. (அ) பரிமேலழகரின் வடமொழிப் புலமையை எடுத்துக்காட்டுக.

(அல்லது)

(ஆ) பரிமேலழகரின் சொல்லாராய்ச்சியைச் சான்றுகள் தந்து விளக்குக.

5. (அ) நாவலர் சோமசுந்தர பாரதியின் உரைச் சிறப்புகளை விளக்குக.

(அல்லது)

(ஆ) சிவக்கவிமணியின் உரைத் திறனை மதிப்பீடு செய்க.